

Master Study Guide: Engineering Mathematics II (SH121)

Unit 1: Linear Algebra & Vector Spaces — [18.7% - 25.2%] of Total Marks

A. Short Answer Questions (SAQs)

1. [2082-Mid] Q1 (5 Marks):

Discuss the meaning of linearly dependent and independent set of vectors in a vector space. Test whether the set of vectors $\{(4, 7, 3), (2, 3, 5), (1, 2, -1)\}$ of $\mathbb{R}^3$ are linearly dependent or independent. (1+4=5)

2. [2081] Q1 (8 Marks):

Discuss the consistency and inconsistency of system of linear equation $AX = B$ with example. Test the consistency of system of equation $x + 2y - z = 3$, $3x - y + 2z = 1$, $2x - 2y + 3z = 2$. Also solve if consistent. (3+5)

3. [2081-Mid] Q2 (8 Marks):

Discuss the meaning of consistency of system of linear equation $AX = B$. Test the consistency of system, $x + y + z = 3$, $2x - y + 3z = 4$, $3x - 5y - z = -3$ and solve if possible using back substitution.

4. [2081-Mid] Q1 (8 Marks):

Define invertible linear transformation with example. Show that the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$, defined by $T(x, y, z) = (x-y, y-z, z)$ is invertible and find formula for T^{-1} . **[REPEATED: Identical to 2081 Q9 Part c]**

5. [2080] Q1 (8 Marks):

Discuss the meaning of consistency and inconsistency of system of linear equation $AX = B$ with example. Test the consistency of system of equation, $3x + 6y + z = 16$, $2x + 4y + 3z = 13$, $x + 3y + 2z = 9$ and solve if possible by back substitution method. (2+4+2)

6. [2079] Q1 (8 Marks):

Define basis and dimension of vector space $V(F)$ with examples. Show that the set of vectors $\{(1, 0, 1), (1, 1, 0), (-1, 0, 1)\}$ forms basis of $\mathbb{R}^3$. (2+6=8)

B. Long Answer Questions (LAQs)

1. [2082-Mid] Q6 (10 Marks):

With example discuss the meaning of linear transformation. Also discuss the meaning of singular, non-singular and invertible linear transformation. If $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$, be defined by $T(x, y, z) = (x-y, y-z, z)^T$, show that T is invertible and also find T^{-1} . **(4+6=10) [REPEATED: Identical to 2081 Q9]**

2. [2081] Q9 (15 Marks):

Define linear transformation and invertible linear transformation with example. Prove that $T: V \rightarrow W$ is invertible iff $\ker T = \{0\}$. Test invertibility of linear map $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x-y, y-z, z)$ and also find T^{-1} if exist. **(2+5+8)**

3. [2080] Q7 (15 Marks):

Define linear transformation and inverse linear transformation with example. Show that a linear mapping $T: V \rightarrow W$ is one to one iff $\ker T = \{0\}$. Also discuss the meaning of invertible linear transformation and mention the criteria for linear transformation to be invertible, show that $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (3x - y, 4x + 2y)$ is invertible. Also find formula for T^{-1} . **(2+4+7+2)**

4. [2079] Q9 (15 Marks):

- **(a)** Define homogeneous and non-homogeneous system of linear equation with example. Test the consistency of system of linear equations, $x + y + z = 4$, $x + y + 2z = 2$, $2x + 2y + z = 5$. **(2+6=8)**
- **(b)** Show that the linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (x, x+y)$ is invertible and also find formula for T^{-1} . **(5+2=7)**

5. [2079] Q11 Part b (8 Marks out of 15):

Define Eigen value and Eigen vector with examples. Find the Eigen values and Eigen vectors of the matrix

$$A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}.$$

(2+6=8) (Note: Part a is categorized under Unit 2)

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Consists of theoretical proofs (such as proving T is invertible iff $\ker T = \{0\}$), definition-based short questions (basis, dimension, consistency definitions), and computational numericals (consistency testing using Gauss elimination, finding inverse transformations T^{-1} , and eigenvalue/vector computation).

- **Direct Repetitions:**

- The linear transformation $T(x, y, z) = (x-y, y-z, z)$ is tested for invertibility and inverse formulation across three papers: **2082-Mid Q6, 2081 Q9, and 2081-Mid Q1.**
- Consistency testing is a guaranteed 8-mark question in every single exam sitting.

- **Key High-Priority Topics:**

- Gauss elimination / rank method for testing system consistency and back substitution.
- Invertibility proofs for linear transformations and the kernel theorem (T is 1-1 iff $\ker T = \{0\}$).
- Basis verification of R^3 set of vectors.
- Characteristic equations for eigenvalues and subsequent eigenvectors.

Unit 2: Three Dimensional Geometry — [24.4% - 25.2%] of Total Marks

A. Short Answer Questions (SAQs)

1. [2082-Mid] Q5 (5 Marks):

Find the equation of sphere having the circle, $x^2 + y^2 + z^2 + 10y - 4z - 8 = 0$, $x + y + z = 3$, as great circle.

2. [2081] Q2 (8 Marks):

Discuss the different forms of straight lines in space. Express the equation of line $x + 2y + 3z - 6 = 0 = 3x + 4y + 5z - 2$ in symmetrical form. **(2+6)**

3. [2080] Q2 (8 Marks):

Find the distance from the point (3, 4, 5) to the point where the line

$$(x - 3)/1 = (y - 4)/2 = (z - 5)/2$$

meets the plane $x + y + z = 2$. **[REPEATED: Identical to 2079 Q2 Part b]**

4. [2079] Q2 (8 Marks):

Prove that the linear equation in x, y, z represents plane surface. Find the distance from the point (3, 4, 5) to the point where the line

$$(x - 3)/1 = (y - 4)/2 = (z - 5)/2$$

meets the plane $x + y + z = 2$. (4+4=8)

B. Long Answer Questions (LAQs)

1. [2081] Q8 (15 Marks):

Define cone and cylinder and also discuss the condition that they reduce into right cone and right cylinder. Find the equation of right circular cylinder having the circle $x^2 + y^2 + z^2 = 9$, $x - y + z = 3$ as its guiding curve. (5+10)

2. [2081] Q10 Part a (8 Marks out of 15):

Find the equation of sphere passing through the points (1, -3, 4), (1, -5, 2) and (1, -3, 0) and whose centre lies on the plane $x + y + z = 0$. (Note: Part b is categorized under Unit 3)

3. [2080] Q8 (15 Marks):

Define three dimensional locus, cone and cylinder and mention the criteria that they reduces to right cone and right cylinder. Deduce the equation of cone having vertex at (α, β, γ) and guiding curve $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, $z = 0$. Also find equation of right cone having vertex at origin and axis $x/1 = y/2 = z/3$, with vertical angle 30° . (2+1+7+5)

4. [2080] Q11 Part b (7 Marks out of 15):

Define sphere and find the equation of sphere passing through points (3, 0, 2), (-1, 1, 1), (2, -5, 4) and having Centre on the plane $2x + 3y + 4z = 6$. (Note: Part a is categorized under Unit 3)

5. [2079] Q8 (15 Marks):

Define cone and cylinder. Deduce the equation of cone with vertex at (α, β, γ) and guiding curve is $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, $z = 0$. Also find the cone with vertex at (3, 2, 1) and base is $2x^2 + 3y^2 = 1$, $z = 1$. (2+8+5=15)

6. [2079] Q11 Part a (7 Marks out of 15):

Prove that the plane $2x - y + 2z - 14 = 0$, touches the sphere, $x^2 + y^2 + z^2 - 4x + 2y - 4 = 0$. Also find the point of contact. (5+2=7) (Note: Part b is categorized under Unit 1)

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** This section demands excellent visualization. It contains derivation of the standard cone equation, right circular cone/cylinder formulation with specified guiding curves, finding equations of spheres passing through three points, tangent plane proofs, and intersection points of lines and planes.
- **Direct Repetitions:**

- The exact distance question: Point (3,4,5), line meets plane $x+y+z=2$ is 100% identical in **2080 Q2** and **2079 Q2**.
- The cone derivation with vertex at (α, β, γ) is repeated in **2080 Q8** and **2079 Q8**.
- **Key High-Priority Topics:**
 - Deduction of the general cone equation with vertex (α, β, γ) .
 - Sphere equation passing through 3 points with a plane containing the center.
 - Condition of tangent plane touching a sphere ($d = r$) and locating the point of contact.
 - Converting unsymmetrical lines to symmetrical forms.
 - Great circle conditions for a sphere.

Unit 3: Multiple Integrals & Curve Parametrization — [19.5% - 25.2%] of Total Marks

A. Short Answer Questions (SAQs)

1. [2082-Mid] Q2 (5 Marks):

Discuss the meaning of parametric representation of a plane curve. If a point $p(x, y)$ moves in a plane such that its position at time t is given by $x = a \cos t$, $y = -b \sin t$, $0 \leq t \leq \pi$. Discuss the particles motion and path traced. (1+4=5) [REPEATED: Similar to 2081 Q3 and 2080 Q3]

2. [2082-Mid] Q3 (5 Marks):

Discuss the geometrical meaning of double integral $\iint_{\tau} f(x, y) dA$. Evaluate

$$\int_0^1 \int_0^{\sqrt{1-x^2}} y^2 dy dx,$$

by reversing the order of integration. (1+4=5)

3. [2081] Q3 (8 Marks):

Define and differentiate between Cartesian, polar and parametric path with example. If a point $p(x, y)$ in plane moves such that $x = a \cos t$, $y = -b \sin t$, $0 \leq t \leq \pi$. Discuss the motion and path traced by the particle. (3+5) [REPEATED: Similar values to 2080 Q3]

4. [2081] Q4 (8 Marks):

Calculate the volume of solid bounded by the surfaces, $z = 0$, $x^2 + y^2 = 1$, $x + y + z = 3$.

5. [2081-Mid] Q3 (8 Marks):

With example define and differentiate between Cartesian, polar and parametric path. If a particle $P(x, y)$ moves in plane such that $x = a \cos t$, $y = -b \sin t$, $0 \leq t \leq \pi/2$, discuss the motion of the particle and trace its path.

6. [2081-Mid] Q4 (8 Marks):

Discuss the meaning of triple integral $\iiint_V f \, dV$, over the region of space. Using triple integral find the volume of sphere $x^2 + y^2 + z^2 = a^2$.

7. [2081-Mid] Q5 (8 Marks):

Evaluate the integral

$$\int_0^a \int_0^{\sqrt{a^2-x^2}} y^2 \sqrt{x^2+y^2} \, dy \, dx,$$

by changing in to polar co-ordinate.

8. [2080] Q3 (8 Marks):

Define the parametric equation with example and discuss why parametric equation are assumed to be more efficient way than Cartesian equation to represent a path. A particle $p(x, y)$ moves in a plane with $x = a \cos t$, $y = -b \sin t$, $0 \leq t \leq \pi$, discuss the motion and path traced by the particle. (1+2+5)

9. [2080] Q4 (8 Marks):

Discuss about the necessity of reducing double integral in Cartesian co-ordinate to polar co-ordinate, with example. Evaluate

$$\int_0^2 \int_0^{\sqrt{2x-x^2}} [xy / \sqrt{x^2+y^2}] \, dy \, dx,$$

by reducing into polar co-ordinate. (2+6)

10. [2079] Q3 (8 Marks):

With example express your views, why parametric system is assumed to be superior to Cartesian system? Describe the motion of a moving particle $p(x, y)$ whose position at time T is given by $x = \cos t$, $y = -\sin t$, $0 \leq t \leq \pi/2$. (3+5=8)

11. [2079] Q4 (8 Marks):

Describe the meaning and application of multiple integral with example. Using the concept of multiple integral find the area of region R : $0 \leq x \leq 2$, $x^2 \leq y \leq 2x$. (3+5=8)

B. Long Answer Questions (LAQs)

1. [2081] Q10 Part b (7 Marks out of 15):

Evaluate the double integral $\iint_R xy \, dx \, dy$, where R is the region bounded by x -axis, the line $y = 2x$ and the parabola $x^2 = 4ay$. (Note: Part a is categorized under Unit 2)

2. [2081] Q11 Part a (7 Marks out of 15):

Using polar integration find the area of region bounded by polar curve $r = a(1 + \cos\theta)$ (printed in exam as $r = (1 + \cos\theta)$). (Note: Part b is categorized under Unit 4)

3. [2080] Q11 Part a (8 Marks out of 15):

Discuss the geometrical meaning of double integral $\iint_C f \, dA$. Evaluate the integral

$$\int_0^1 \int_0^{\sqrt{1-x^2}} y^2 \, dx \, dy,$$

by changing the order of integration. (Note: Part b is categorized under Unit 2)

4. [2079] Q10 (15 Marks):

- (a) Evaluate the integral

$$\int_0^a \int_0^{\sqrt{a^2-x^2}} \sqrt{x^2+y^2} \, dy \, dx,$$

by changing into polar co-ordinate. (8)

- (b) Describe the meaning of $\int_{\theta=\alpha}^{\theta=\beta} f(\theta) \, d\theta$. Using the same concept find the area of region that lies inside the circle $r = 1$ and outside cardioid $r = 1 - \cos\theta$. (2+5=7)

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Integrates theoretical conceptualization (geometrical meaning of double/triple integrals, parametric system superiority) and solid computational math (polar conversions, reversing order of integration, and finding bounded area/volumes).
- **Direct Repetitions:**
 - Parametric curve discussion of elliptical arcs ($x = a \cos t$, $y = -b \sin t$) is extremely popular, appearing in **2082-Mid Q2**, **2081 Q3**, **2081-Mid Q3**, and **2080 Q3**.
 - Reversing the order of integration of $\int y^2 \, dy \, dx$ over a circular quadrant is shared between **2082-Mid Q3** and **2080 Q11(a)**.
- **Key High-Priority Topics:**
 - Parametric path motion analysis and curve tracing.
 - Cartesian to polar coordinate transformations for double integrals (particularly regions bounded by circles $x^2+y^2=a^2$ or $x^2+y^2=2ax$).
 - Changing the order of double integration with variable limits.

Unit 4: First Order Ordinary Differential Equations — [6.5% - 13.0%] of Total Marks

A. Short Answer Questions (SAQs)

1. [2082-Mid] Q4 (5 Marks):

Discuss the meaning of differential equation and meaning of its solution with example. Find the differential equation corresponding to the family of curve $y = Ae^x + Be^{-x}$. **(2+3=5)**

[REPEATED: Similar to 2080 Q5 Part c]

2. [2081] Q5 (8 Marks):

Define first order and first degree differential equation with example and discuss different techniques of solving it. Using one of the methods solve, $(x^3 + 1) dy/dx + 3x^2y = \sin^2x$. **(3+5)**

[REPEATED: Identical to 2080 Q6 Part b]

3. [2081-Mid] Q6 (8 Marks):

Define meaning of order and degree of differential equation. Solve the equation, $dy/dx - y \tan x = -y^2 \sec x$. (Note: This is Bernoulli's form of DE).

4. [2080] Q5 (8 Marks):

Discuss the meaning of solution of differential equation with example and also discuss the process of formation of differential equation corresponding to family of curve, $f(x, y, c_1, c_2) = 0$. Form the differential equation corresponding to family of curve, $xy = Ae^x + Be^{-x}$. **(2+2+4)**

5. [2080] Q6 (8 Marks):

Define linear differential equation of first order and first degree with example. Solve the differential equation, $(x^3 + 1) dy/dx + 3x^2y = \sin^2x$. **(2+6)**

6. [2079] Q5 (8 Marks):

With suitable example define the linear homogeneous differential equation of first order. Solve the differential equation $(4x + 6y + 5) dy - (3y + 2x + 4) dx = 0$. **(2+6=8)** (Note: This is reducible to homogeneous form).

B. Long Answer Questions (LAQs)

1. [2081] Q11 Part b (8 Marks out of 15):

Discuss the process of forming differential equation corresponding to given family of curve. Find the differential equation corresponding to family of curve $y = ax^3 + bx^2$. (Note: Part a is categorized under Unit 3)

Unit 5: Higher Order Ordinary Differential Equations — [6.5% - 12.2%] of Total Marks

A. Short Answer Questions (SAQs)

1. [2081] Q6 (8 Marks):

Define complementary function and particular integral with example. Solve $(D^2 + 1)y = x^2 \sin 2x$. (2+6)

2. [2079] Q6 (8 Marks):

Find the complete solution of differential equation $d^2y/dx^2 - 4 dy/dx + 4y = x^2 + e^{2x}$. (Note: The exam paper contains a typo reading "+ 4 = x^2", which mathematically must be "+ 4y = x^2" as confirmed in the identical 2080 Q10). [REPEATED: Identical to 2080 Q10 Part c]

B. Long Answer Questions (LAQs)

1. [2080] Q10 (15 Marks):

Define the nth order differential equation with constant coefficient. Also describe the process of finding solution of differential equation $d^2y/dx^2 + p_1 dy/dx + p_2y = Q(x)$. Show that if $y = f_1(x)$ and $y = f_2(x)$ are two independent solutions of $d^2y/dx^2 + p_1 dy/dx + p_2y = 0$, then $y = Af_1(x) + Bf_2(x)$ is also solution of same equation. Find the solution of $(D^2 - 4D + 4)Y = x^2 + e^{2x}$. (2+4+9)

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Demands linear higher-order differential equation proofs (superposition principle) and heavy computational procedures for finding Complementary Functions (CF) via auxiliary equations and Particular Integrals (PI) using operator methods $(1/f(D) \cdot Q(x))$.
- **Direct Repetitions:**
 - The complete numerical solution of $(D^2 - 4D + 4)y = x^2 + e^{2x}$ is 100% duplicated in **2080 Q10** and **2079 Q6** (rectified).
- **Key High-Priority Topics:**
 - Finding CF when roots are real, equal, or imaginary.
 - Operator methods for finding PI when $Q(x) = e^{ax}$, $Q(x) = x^m$, and $Q(x) = e^{ax}V$ or xV (specifically $x^2 \sin 2x$).
 - Linear superposition principle theorem proof ($y = Af_1 + Bf_2$ is a solution).

Unit 6: Series Solutions of Differential Equations — [12.2% - 12.2%] of Total Marks

A. Short Answer Questions (SAQs)

(No short answer questions have been historically set for this unit; series solutions are exclusively allocated as major 10-mark or 15-mark long questions).

B. Long Answer Questions (LAQs)

1. [2082-Mid] Q7 (10 Marks):

Define the meaning of singular point of differential equation $p_0(x) \frac{d^2y}{dx^2} + p_1(x) \frac{dy}{dx} + p_2(x)y = 0$. Find the power series solution of $3xy'' + 2y' + y = 0$, about $x = 0$. **(2+8=10)**

2. [2081] Q7 (15 Marks):

With particular example verify that power series solution of differential equation $p_0 \frac{d^2y}{dx^2} + p_1 \frac{dy}{dx} + p_2y = 0$ exist. Also define ordinary and singular point of $p_0 \frac{d^2y}{dx^2} + p_1 \frac{dy}{dx} + p_2y = 0$ and find series solution of $(1 - x^2)\frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0$, about $x = 0$. **(3+2+10)**

3. [2080] Q9 (15 Marks):

Define regular and irregular singular point of the differential equation $p_0 \frac{d^2y}{dx^2} + p_1 \frac{dy}{dx} + p_2y = 0$. Find the regular and irregular singular points of $x^2(x-2)^2 \frac{d^2y}{dx^2} + 2(x-2) \frac{dy}{dx} + (x+3)y = 0$. Also find the series solution of $x(x-1)\frac{d^2y}{dx^2} + (3x-1)\frac{dy}{dx} + y = 0$, about $x = 0$.

[REPEATED: Identical series solution part to 2079 Q7]

4. [2079] Q7 (15 Marks):

Define ordinary point and singular point of differential equation $p_0(x) \frac{d^2y}{dx^2} + p_1(x) \frac{dy}{dx} + p_2(x)y = 0$. Identify the regular singular point of differential equation, $x(x-1)\frac{d^2y}{dx^2} + (3x-1)\frac{dy}{dx} + y = 0$. Also find power series solution of the differential equation about $x = 0$. **(3+2+10=15)**

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Represents the highest mathematical difficulty in the syllabus. It contains rigorous conceptual definitions (ordinary, singular, regular, and irregular points), mathematical proof of series existence, and the lengthy Frobenius series method (solving about $x = 0$).
- **Direct Repetitions:**
 - The differential equation $x(x-1)y'' + (3x-1)y' + y = 0$ is solved for power series about $x = 0$ in both **2080 Q9** and **2079 Q7**.
- **Key High-Priority Topics:**
 - The Frobenius power series method (assuming $y = \sum c_r x^{m+r}$, finding the indicial equation, and recurrence relations).

Subject Priority Planner (ROI-Based Strategy)

The table below ranks all chapters from the highest return-on-investment (highest priority for scoring/passing) to the lowest based on exam weightage and repeatability:

Rank	Unit No.	Unit Name	Priority Class	Historical Mark Percentage	Core Focus Areas to Pass/Score
1	Unit 2	Three Dimensional Geometry	Highest	24.4% - 25.2%	Right circular cones, cylinder equations with circular guides, plane meeting line distance, great circles on spheres.
2	Unit 3	Multiple Integrals & Curve Parametrization	Highest	19.5% - 25.2%	Cartesian-polar double integrals, changing integration order, ellipsoid/cardioid parametric curve analysis, area/volume.
3	Unit 1	Linear Algebra & Vector Spaces	Highest	18.7% - 25.2%	System consistency (Gauss elimination), linear transformation invertibility and T^{-1} formulation, eigenvalues & vectors.
4	Unit 6	Series Solutions of DEs	High	12.2% - 12.2%	Regular/irregular singular points definition, Frobenius power series method around $x = 0$, Legendre series solution.
5	Unit 4	First Order Ordinary DEs	Medium	6.5% - 13.0%	Reducible-to-homogeneous ODEs, integrating factor linear ODEs, Bernoulli equations, DE formation by eliminating constants.
6	Unit 5	Higher Order Ordinary DEs	Medium	6.5% - 12.2%	Superposition theorem proof, finding complementary function (auxiliary equations), operator methods for particular integrals.